



CUSTOM VISION / TTS CURRENCY RECOGNITION SYSTEM FOR THE BLIND

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ABSTRACT

Facing multiple challenging problems such as the malmanagement of the public health sector and its industrialization, Egypt's situation fueled the search for an ICT-based solution to assist the disabled and abord their marginalization, targeting the blind/visually impaired population. This study introduces an AI-driven Currency Recognition System (CRS) utilizing the custom vision service API (CV API) interpretation and text-to-speech (TTS) services, addressing challenges faced by visually impaired individuals in handling currency transactions and calculations. Initially, this ergonomic aiming to distinguish and identify Egyptian banknotes through image processing techniques and feature extraction, was chosen after comparisons with the SIFT application and color feature extraction which proved this device's effectiveness in the accuracy of identifying the denomination of EGP count and calculation ability for the blind, fulfilling the design requirements. The proposed method displays a notable accuracy rate of 0.9 approximately, at a prior comparison of 0.71, and combined summation of banknotes at a swift runtime of 1.05 seconds. The ergonomic was tested under simulated conditions for the blind. On this basis, large-scale applications of the process were estimated and the potential of the technology replacing existing ones was concluded.

INTRODUCTION

Egypt is currently battling with challenges in its quest for public health industrialization. Issues such as underfunding, subpar healthcare quality, and a lack of essential technical equipment have posed significant obstacles. However, people with disabilities face even greater health disparities, attributable not only to their primary medical conditions but also to the ramifications of social and industrial marginalization. In a nation with an estimated 3 million visually impaired citizens, the imperative for enhancing assistive technologies to safeguard this population and combat financial fraud is undeniable.

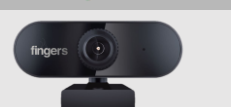
The specific problem at hand is to address the difficulties faced by people with disabilities, particularly the visually impaired, by harnessing the potential of Artificial Intelligence (AI). One of the pressing challenges for the blind community is the issue of financial scams and their inability to identify banknotes accurately. This necessitates the development of a currency recognition system (CRS) that can reliably determine the value of Egyptian banknotes (EGP), regardless of factors like rotation, lighting conditions, scaling, or any other factors that may degrade the quality of the currency, such as wrinkles, noise, or stripes.

While there is an ongoing effort to identify the most effective technologies to aid visually impaired individuals in financial transactions, existing solutions have their limitations. For instance, a mobile application has been developed that employs the SIFT algorithm to recognize and catalog currency, offering advantages in foreign banknote detection but struggling when faced with folded or creased banknotes. Another system focuses on identifying counterfeit currency with accessibility features but is hindered by its relatively slow scanning rate.

Yet, the pursuit of the optimal technology in the Egyptian context remains an unresolved matter. This research aims to implement a robust CRS service with above-threshold accuracy and calculational change in ability as design requirements. A modern-day approach featuring a glove module equipped with deep learning capabilities, attachable to the palm, and powered by computer vision interpretation, is further refined through training using a dedicated algorithm for currency denomination recognition promises to meet the specified design requirements.

MATERIALS

Table.1.(Materials used)

Material	Quantity	Cost L.E.	Illustration
Raspberry Pi 3 B+	1	---	 Fig.(1)
Webcam camera	1	900	 Fig.(2)
4 Push Button	1	2	 Fig.(3)
Earphone	1	35	 Fig.(4)
Memory card	1	---	 Fig.(5)
Power bank	1	300	 Fig.(6)
Total cost		1237	

METHODS

Hardware methods (1) Insert the memory card into the Raspberry camera module (2) Connect Raspberry Pi to web camera module (3) Connect (4-pin push) button to Raspberry Pi 3 (4) Attach the module consisting of Raspberry Pi and camera module to the wool glove to provide blind easy transportation. (5) Connect the Raspberry Pi to the power bank with long USB cable.

Software methods This code :<https://docs.google.com/document/d/1R4ts06wuxC8C8XMBagTwrt1bJdyfDM5hP1f1aO6ss/edit?usp=sharing> involves tasks related to computer vision, audio processing, and API interaction: It captures frames from Raspberry Cam, performs object detection using the Custom Vision service from Microsoft Azure, plays audio feedback, and manages monetary transactions based on the detected objects.

1)Training:data set of 800 self-taken pictures (100/banknote) trained on Microsoft Azure.

2) importing libraries: (os, cv2, requests, threading, time, json, gTTS, play sound)

3)variables and push button-define variables (api_endpoint, api_key, headers, etc.) and methods (talk, execute_prediction, capture_frames) -allows the user to trigger object detection by pushing the button (one in a sec to capture, 'double in one sec' to inquire about total money)

4) Text-to-speech function: It utilizes the gTTS library to convert the given text (mytext) into speech. calling the talk function with the text "الصورة التقطت" which translates to "The image is captured" in Arabic.

5)Displaying Raspberry interface on the VNC viewer

Table.2.(ICT processes)	
Sensing	Camera module
Communicating	Raspberry pi
Data analysis	Azure training
Controlling	Button
Decision	Summation-detection

Fig(7). Data set samples.

Table.3.(Computer vision components)

Data augmentation	Rotation , flipping ,scaling and noise backgrounds for banknote pictures.
Model fitting	Azure is trained to classify currency by patterns of side value and colors.
Training convolution	By CNN , math operations to extract characteristics and geometrical patterns
Pooling	Model reduces complexity and becomes more sensitive to spatial variations.
Validation	Testing new data set to evaluate trained model performance

Test plan
Test 1: System's accuracy. 20 frames were taken for each banknote (10 frames –the Arabic side and 10 frames –the English side). The model was run with a set accuracy threshold of 0.7. The accuracy of each side was calculated.

Test 2:Operational time/calculation change in ability: Simulation of blind parameters was conducted. Using the module, capture frames for two banknotes. The model then calculates the time needed for the summation of the frames and procedures for 3, 4,5,6,7, and 8 banknotes were repeated.

RESULTS

Table.4.(Calculation ability and operational time)

No. of papers summed	Correct yield	Operational time (seconds)
2	Yes	0.927276± 0.01
3	Yes	0.993782± 0.01
4	Yes	1.017829± 0.01
5	Yes	1.142829± 0.01
6	Yes	1.007383± 0.01
7	Yes	0.998874± 0.01
8	Yes	1.284839± 0.01
Average		1.053258± 0.01

Test plan 2

The following table (4) depicts the summation operation for the banknotes (200, 100, 50, 20-new, 20-old, 10-new, 10-old, and 5) according to the number of papers summed, whether the summation gives the correct yield and the operational time. The model exhibited an average operational time of (1.053258 seconds).

Table.5.(Detection accuracy)

Denomination value in EGP	5	10	10 new	20	20 new	50	100	200
Average confidence level (Arabic side)	0.89614	0.94445	0.95072	0.954618	0.96314	0.8999	0.88617	0.91051
Average confidence level (English side)	0.93196	0.94541	0.9315	0.92127	0.95157	0.8949	0.90882	0.91472

Fig(9). Sample of accuracy.

Frame captured and saved as image.jpg
Probability: 0.97802174, TagName: 20_new
Frame captured and saved as image.jpg
Probability: 0.98458123, TagName: 20_new
Frame captured and saved as image.jpg
Probability: 0.967668, TagName: 20_new
Frame captured and saved as image.jpg
Probability: 0.9690658, TagName: 20_new

ANALYSIS

Grand challenges presented in increasing the industrial bases in Egypt and working to eradicate public health issues are confronted by this Egyptian banknote detector wearable glove which improves the daily interactions and the well-being of the blind and visually impaired people with society by allowing them to know the banknote value besides calculating the total summation of the money present. The utilization of Raspberry Pi and other electronics will result in an expansion in the industrial manufacture of these parts to meet demand.

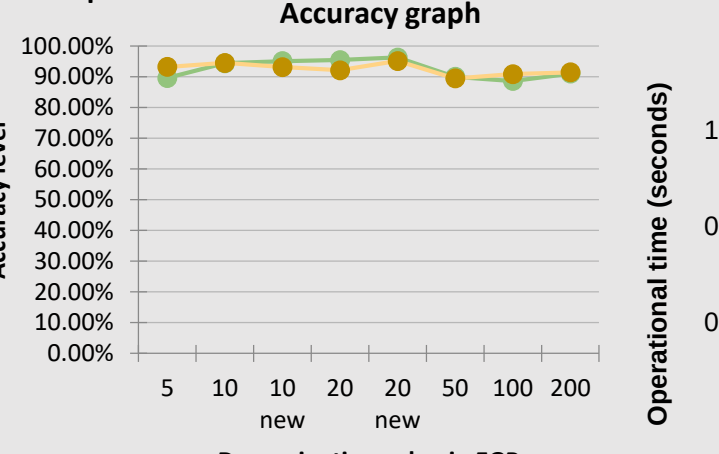
Design requirements were fulfilled by the solution as:

High detection accuracy was achieved to be an average that ranges from 88.6% to 96.3% after the addition of a threshold that equals (0.7) to the code.

The independent blind person was attained due to 99.99% accuracy for the calculation performance for the amount of money present with an average operational time that equals 1.05 sec.

Positive results

- The average confidence level in denomination detection falls within the impressive range of 88.617% to 96.14% as shown in fig (10).
- The operational time, averaging (1.05 sec) as shown in fig (11), is excellent and it operates smoothly without encountering any delays.
- The adoption of no algorithm for model training proved to be a wise decision, given its exceptional accuracy in detection and swift response rate.



Fig(10). accuracy graph.

Prior Solution comparison Table.6.(comparison with prior solution)

Parameter	Selected solution	Currency recognition using a smartphone
Calculation process	Available	Unavailable
Hardware	Present	Software only
Programing language	Python	Java
Algorithm	No algorithm	SIFT
Average accuracy	From 88.6% to 96.3%	71%
Average Operational time (s)	1.05	72.9

Large-scale application

In Egypt, the large-scale would find an optimal execution in densely populated urban centers such as Cairo and Alexandria. These locations offer higher foot traffic and serve as central hubs, increasing accessibility for individuals with visual impairments across various socioeconomic backgrounds. The widespread deployment within these urban areas addresses the needs of a larger population and provides inclusive access to financial services. While the small-scale implementation for 4,000 EGP would cover limited resources, scaling it up to serve a broader population across Egypt would require an estimate of about 4 billion EGP.

Laws and theories

Accuracy= $\frac{\text{Numbers of succes tests}}{\text{Numbers of success tests} + \text{Numbers of failure tests}} \times 100$

Arithmetic Mean (Average) = $\frac{\sum \text{Values of the set}}{\text{total numbers of values in the set}}$

Combination: C = $\frac{n!}{k!(n-k)!}$

Learning transfer

BI.3.01: (Describe the relationship between the structure and the function of a neuron in its role in cell communication.)Understanding the function of neuron parts provided an illustration of how smart systems operate, with connections and communication across system components enabling optimal decision-making and situational control.

CH.3.01: (Differentiate: law and theory; theory and experiment; qualitative and quantitative; and hypothesis and theory) By reading scientific papers and examining their technique, we were able to conduct our research about ICT and its applications in assisting disabled people with the assistance of knowing the difference between theory and experiment.

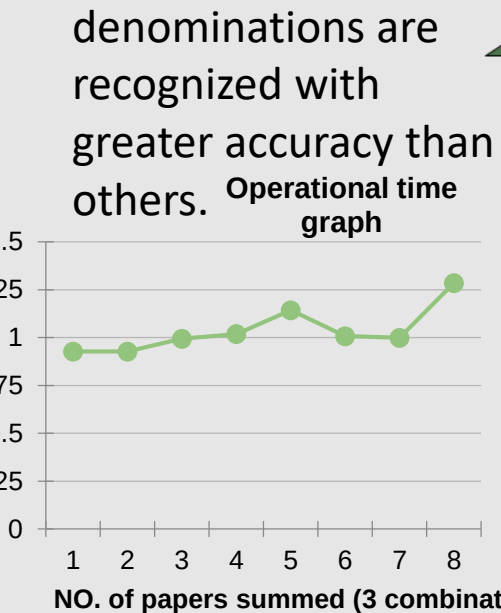
CS.3.02: (Demonstrate understanding of the requirements for creating a network) Given that networking is a fundamental component of ICT, knowledge of the prerequisite hardware, software, internet connection, and database was necessary to comprehend the concept. It also aided in the formation of our project.

PH.3.04: (Design a system of communication through applying previous knowledge) This learning outcome gave us the necessary knowledge about communication and its fundamental elements, such as a transmitter, a medium, a receiver, and so forth.

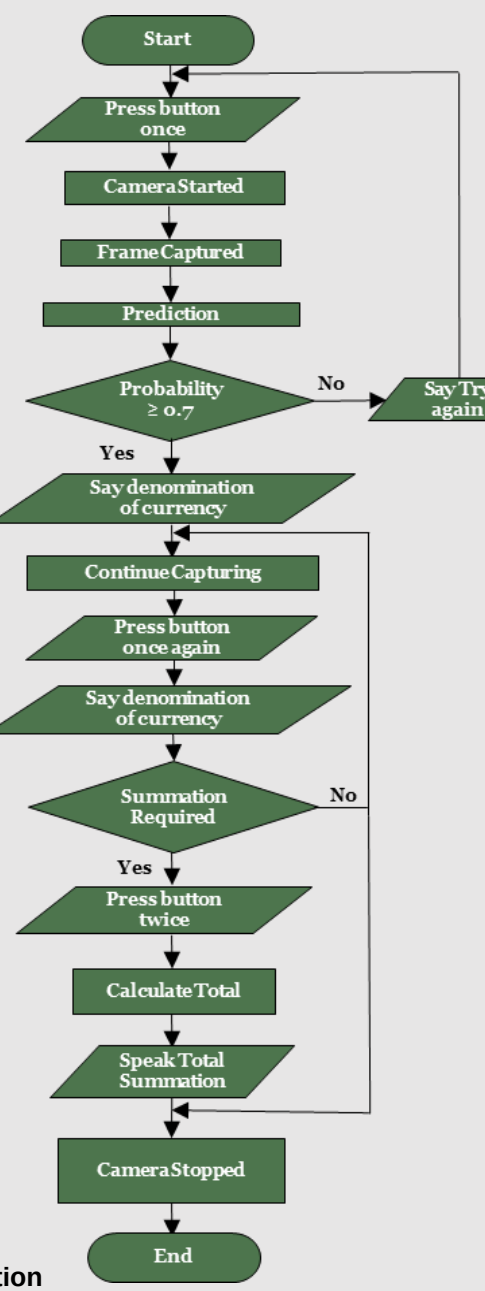
MA.3.02 : (Analyze the behavior of functions by investigating critical points using first and second derivatives) Local maxima and local minima were used to describe the operational time graph by determining the maximum operational time (1.284839 seconds) and the minimum operational time (0.927276 seconds).

Negative results

- Incorrect recognition denominations, the final summation value will be inaccurate. And the system's performance will be suboptimal when using low-quality cameras.
- Certain sides of the denominations are recognized with greater accuracy than others.



Fig(11). Operational time graph.



Fig(12). Project flowchart.

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CONCLUSION

The main objective of this study was to detect Egyptian banknotes using a wearable glove equipped with a Raspberry Pi module and a model trained on Azure to denominate (200, 100, 50, 20-new, 20-old, 10-new, 10-old, and 5) for the Blind and visually impaired. With both design requirements met optimally, the model demonstrated positive results for average confidence level in denominations detection within 88.617% to 96.314%, as well as calculation code that demonstrated no incorrect summation and an operational time averaging (1.05 sec), Challenging aspects included the degradation of the system's performance when using cameras of low-quality and reduced accuracy for specific currency sides. In contrast to prior solutions such as employing a smartphone for money recognition for blind and fake currency detection using color features extraction, the proposed solution appears more promising due to its high accuracy and quick operating time. Large-scale applications were selected in order to scale it up to service a larger population in Egypt, which is expected to cost 4 billion Egyptian pounds. The results of the study showed that Egypt's goals of expanding its industrial base and eradicating public health problems had been met.

RECOMMENDATION

- Instead of employing the Webcam, consider replacement with the Raspberry Pi camera module V2 (8 MP, 1080p) or a higher-quality model.
- To keep the blind or visually impaired up-to-date and more at ease when using the project, update the dataset frequently with the designs of new banknotes.
- One EGP banknote should be added to the training model of a more advanced and accurate system. Currently, it is not added since its value is not as high as other banknotes' values, and its addition will result in a decline in the accuracy of currency detection.
- Expand the dataset by incorporating additional images of banknotes captured from diverse angles, under varying lighting conditions, and exhibiting characteristics like wrinkles. This will enhance the average confidence level and improve the overall detection performance.

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